

QueryQuest Assignment 1

Database Management Systems

June 7, 2026

Part I: Short Answer Questions

Q1. A DBMS is responsible for the storage of data in the form of structured models (like tables, graphs, key-value pairs etc). It is different from a traditional file system as follows:

- It can easily track who made changes to the file and allows simultaneous updation of data without data corruption or system crashing.
- It only allows specific authorised users to change the data, thus ensuring Data Security.
- It reduces data redundancy by eliminating duplicated files.

Consider a scenario where users are booking tickets for a movie using an app. Clearly there is a possibility of simultaneous booking and if traditional file system was used then there could be a possibility of double-booking or system crashes. On the other hand, DBMS ensures that one person's booking is finished before the others preserving data integrity.

Q2. In an ER diagram, a weak entity is represented by double rectangle whereas a strong entity is represented by a single rectangle. A strong entity has a primary key that uniquely identifies each tuple but on the other hand, a weak entity is dependent on the strong entity for uniquely identifying its rows. The primary key of a weak entity is formed by combining the primary key of its identifying strong entity with its own partial key. This relationship is called an identifying relationship and is represented by double diamond.

Q3. Clearly the phone number of students should be modeled as a multivalued attribute since a student can have multiple phone numbers and thus this attribute could have multiple data entries.

Q4. A candidate key is a minimal set of attributes that can uniquely identify a tuple. A super key is a superset of candidate key with redundant attributes which aren't important for uniqueness.

In the above relation, roll_no and email are the two candidate keys as they uniquely identify each record & the set {roll_no, branch} and {email, name} are the examples of super keys.

Q5. The constraint that is violated in this case is Referential Integrity constraint.

The Referential Integrity constraint ensures that foreign keys must hold values either from the primary key of the referenced table or remain NULL.

The database engine will actively abort the insertion operation and throw an error.

Q6. A separate relation known as a relationship table (also called a junction) must be created.

Its primary key is a composite primary key consisting of the combination of the primary keys of both participating entity relations.

Because both sides of the relationship exhibit a maximum cardinality of 'many'. If we attempt to represent the link as a column in entity Table A, a single row in Table A would need to link to multiple rows in Table

B. This would either force a single cell to store a multi-valued list or force the duplication of rows in Table A.

Part II: Long Answer Questions

Clearly, according to the scenario, there are 4 entities namely Customers, Orders, Restaurants and menu_items. Their key attributes are customer_ID, order_ID, rest_ID and item_ID.

We have the following relationships:

- Customer & Orders - 1:N (A single customer can place multiple orders over time, but each individual order receipt belongs strictly to exactly one customer.)
- Orders & Restaurants - N:1 (A restaurant can receive and process multiple orders from the platform, but any single order receipt is routed to exactly one restaurant.)
- Orders & Menu_items - M:N (An order can contain multiple different menu items, and a single menu item like "Chicken Biryani" can be selected across many different orders.)
- Restaurants & Menu_items - 1:N (A restaurant can host many different dishes on its menu, while each specific menu item record is uniquely owned and priced by exactly one restaurant.)

Here, phone number is an example of a multi-valued attribute as a single customer can have multiple phone numbers.

Also address could be the composite attribute as it can be broken down to individual attributes like pincode, state etc.

Relational schema for Order and MenuItem junction table:

Relational Schema Notation:

Order_Details(order_ID, item_ID, quantity)

Component Details:

- **Primary Key:** The composite primary key consists of (order_ID, item_ID). This unique combination ensures that the exact same menu item cannot be added as a duplicate row within a single order.
- **Foreign Keys:**
 - order_ID is a Foreign Key referencing the order_ID primary key of the **Orders** table.
 - item_ID is a Foreign Key referencing the item_ID primary key of the **Menu_Items** table.
- **Additional Attributes Required:** quantity
The **quantity** attribute must be placed here because it is a descriptive attribute of the relationship itself. It tracks the exact number of units ordered for a specific item on a specific receipt, which cannot belong to either parent table independently.

Question 2: Design Justification

The Doctor–Patient relationship is modeled as Many-to-Many (M:N) because a patient can be treated by multiple specialist doctors during their hospital stay, and a single doctor simultaneously manages multiple patients. In the relational schema, this junction creates an associative entity table named **Treats**. Its composite primary key consists of (Doctor_ID, Patient_ID), elegantly combining the primary keys of both strong entities to map the links uniquely.

Regarding structural participation, the **Patients** entity has total participation in both the **admitted** and **Treats** relationships. This is because a person cannot exist in an active hospital patient tracking system

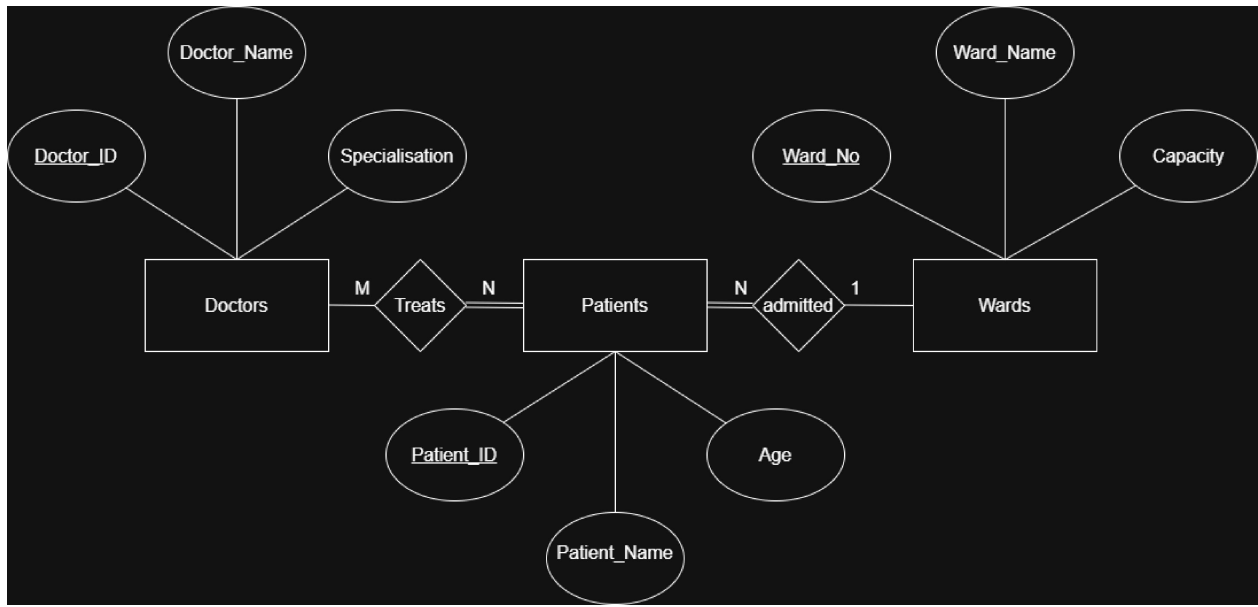


Figure 1: Question 1: Hospital ER Diagram

without being assigned a physical bed in a **Ward** and being under the active care of at least one **Doctor**. Conversely, **Doctors** and **Wards** have partial participation, as a new ward might be empty temporarily, or a consulting doctor might not have an assigned inpatient at a given moment.